Keldysh Ionization Based-FDTD Modeling of Laser-Induced Damage Threshold of MLD-IBS Compression Gratings for Petawatt 2 µm Laser systems

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[2] *Department of Electrical and Computer Engineering, Colorado State University, Fort Collins, Colorado 80523, USA*
[3] *XUV Lasers Inc., Fort Collins, CO 80525, USA*
[4] *Department of Electrical and Computer Engineering, The Ohio State University, Columbus, OH 43210, USA*
[5] *Department of Physics, The Ohio State University, Columbus, OH 43210, USA, OH 43210*

Acknowledgements: **This work was supported by the DOE STTR grant no. DE-SC0019900 and the DOE ARDAP award DE-SC0025577 and used resources from the Ohio Supercomputer Center (OSC).**

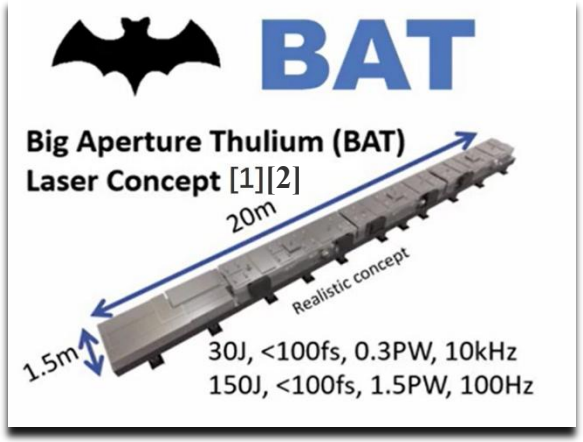
Motivation

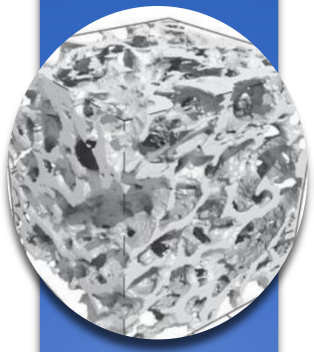
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Real-world laser-driven accelerators, x-ray and ion sources need **powerful, efficient short-pulse lasers** with **high peak and average power**.

- Pumped by diode laser
- Long radiative lifetime ~15 ms
- High wall-plug efficiency
- PW power at 1–100 kHz repetition rate

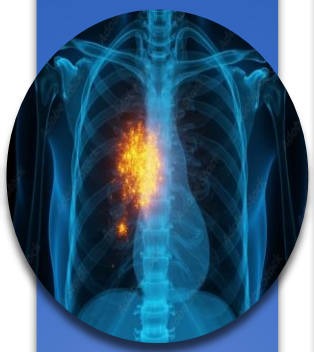
Emission at ~2 μm




X-ray Microscopy


EUV Lithography


Particle Accelerator


Cancer Treatment

……

X-ray → Electron → Ion → ……

[1] Issa Tamer, et al., "Demonstration of a compact, multi-joule, diode-pumped Tm:YLF laser," Opt. Lett. 46, 5096-5099 (2021)
[2] Issa Tamer, et al., "1 GW peak power and 100 J pulsed operation of a diode-pumped Tm:YLF laser," Opt. Express 30, 46336-46343 (2022)

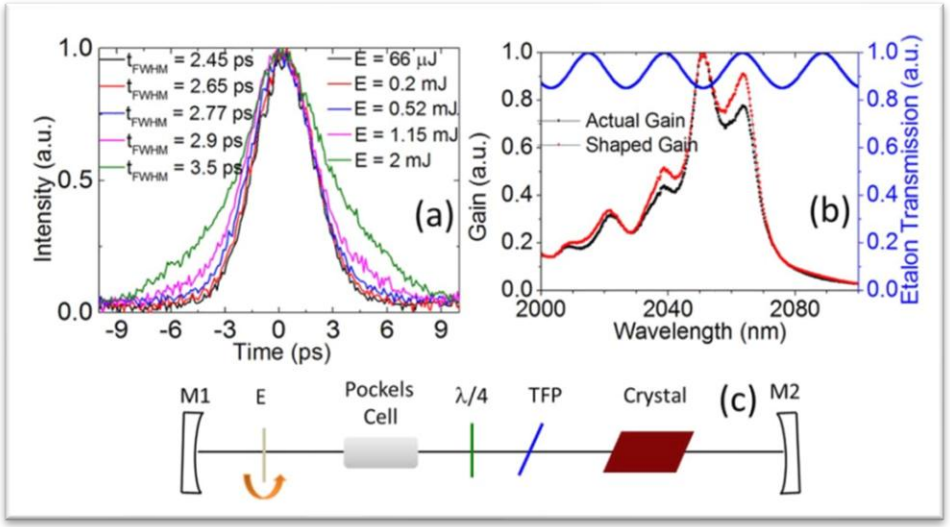
Motivation

Emission at ~2 μm

1 kHz repetition rate

Narrowing bandwidth to have ~2.4 ps

Ho:YLF Regenerative Amplifiers [3]

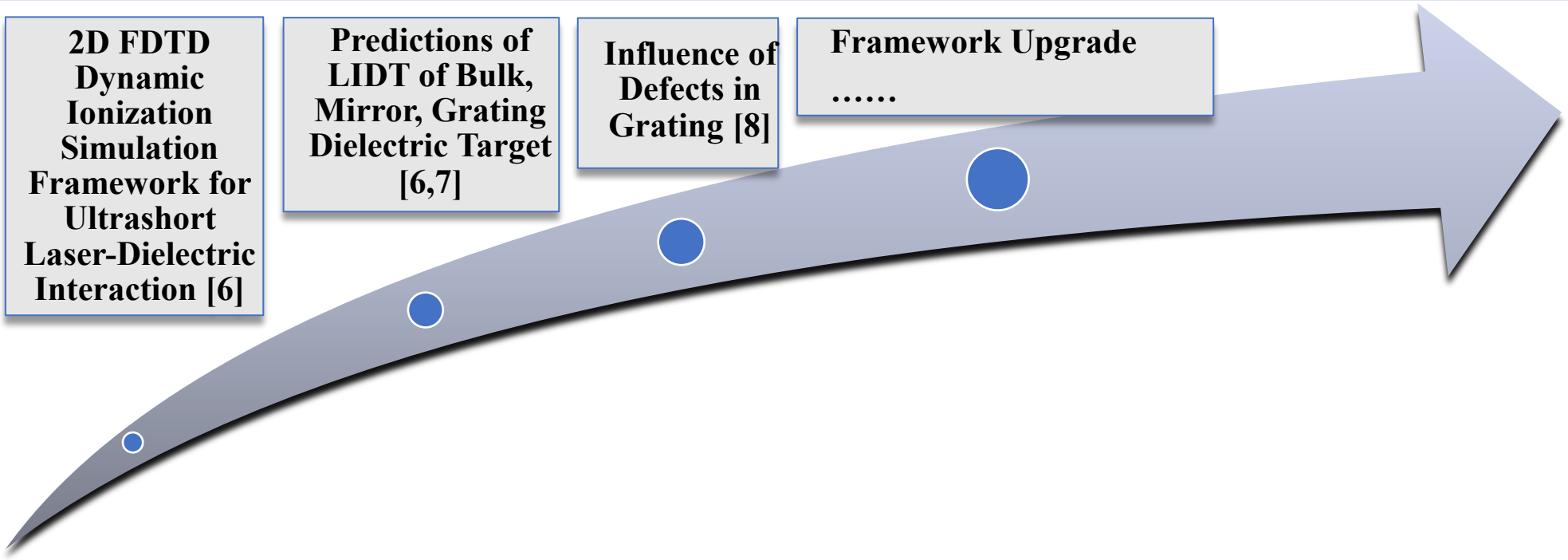


- My advisor Dr. Enam is also working on ARDAP Project for compressor and 2 μm CPA laser system front end development

- Failure of optical components is a key impediment of existing CPA-based ultrafast, high-energy laser systems [4].
- Gratings are identified as one of the weakest parts in the CPA system [5].
- Crucial to increase the Laser Induced Damage Threshold (LIDT).

[3] Krishna Murari, et al., "Intracavity gain shaping in millijoule-level, high gain Ho:YLF regenerative amplifiers," Opt. Lett. 41, 1114-1117 (2016)
[4] Matthias F. Kling, et al., Basic Research Needs Workshop on Laser Technology Report of the Department of Energy Office of Science Workshop August 15-17, 2023 (U.S. Department of Energy Office of Science, January 5, 2024), iii, 4, 44, 49-51, 54, etc.
[5] Kiani, et al., "High Average Power Ultrafast Laser Technologies for Driving Future Advanced Accelerators" (technical report), Journal of Instrumentation 18, no. 08 (2023): 9

Our Efforts



[6] S. Zhang, C. Menoni, V. Gruzdev, and E. Chowdhury, "Ultrafast laser material damage simulation—a new look at an old problem," Nanomaterials 12, 1259 (2022)
[7] S. Zhang, et al., "Femtosecond damage experiments and modeling of broadband mid-infrared dielectric diffraction gratings," Opt. Express 29, 39983-39999 (2021)
[8] S. Zhang, Z. Su, C. S. Menoni, and E. A. Chowdhury, "Influence of defects on the femtosecond laser damage resistance of multilayer dielectric gratings," Opt. Lett., vol. 48, no. 5, pp. 1212–1215, Mar. 2023. DOI: 10.1364/OL.483581.

Our Efforts

2D FDTD Dynamic Ionization Simulation Framework for Ultrashort Laser-Dielectric Interaction [6]

Predictions of LIDT of Bulk, Mirror, Grating Dielectric Target [6,7]

Influence of Defects in Grating [8]

Framework Upgrade ……

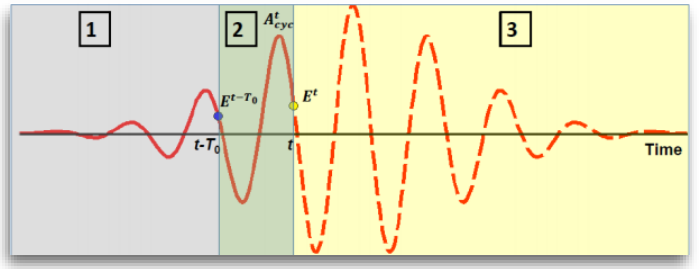
FDTD Field Solver

Keldysh photoionization[9] & Impact Ionization[10]

e-i & e-n with Conditional Hard Collisions[11] & Heating

Drude Modification on Permittivity

Cycle-averaged approach [6]



A sketch of time profile of electric field at a fixed space point

[6] S. Zhang, C. Menoni, V. Gruzdev, and E. Chowdhury, "Ultrafast laser material damage simulation—a new look at an old problem," Nanomaterials 12, 1259 (2022)
[9] L. V. Keldysh, "Ionization in the field of a strong electromagnetic wave," J. Exp. Theor.Phys., vol. 20, no. 5, pp. 1307–1314, 1965
[10] L. V. Keldysh, "Kinetic theory of impact ionization in semiconductors," SOVIET PHYSICS JETP, vol. 37, no. 10, 1960.
[11] Callen, J.D. Fundamentals of Plasma Physics; University of Wisconsin Press: Madison, WI, USA, 2006.

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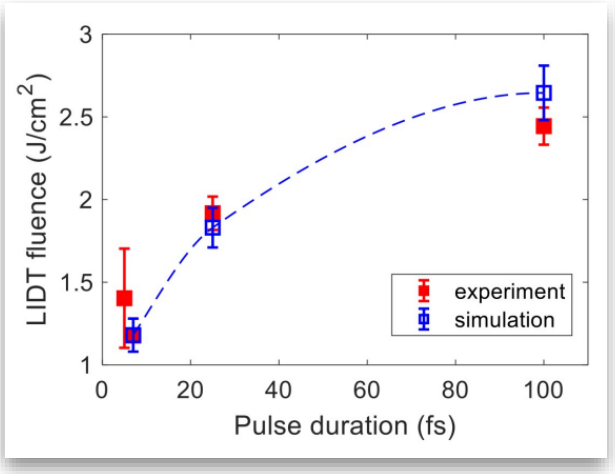
2D FDTD Dynamic Ionization Simulation Framework for Ultrashort Laser-Dielectric Interaction [6]

Predictions of LIDT of Bulk, Mirror, Grating Dielectric Target [6,7]

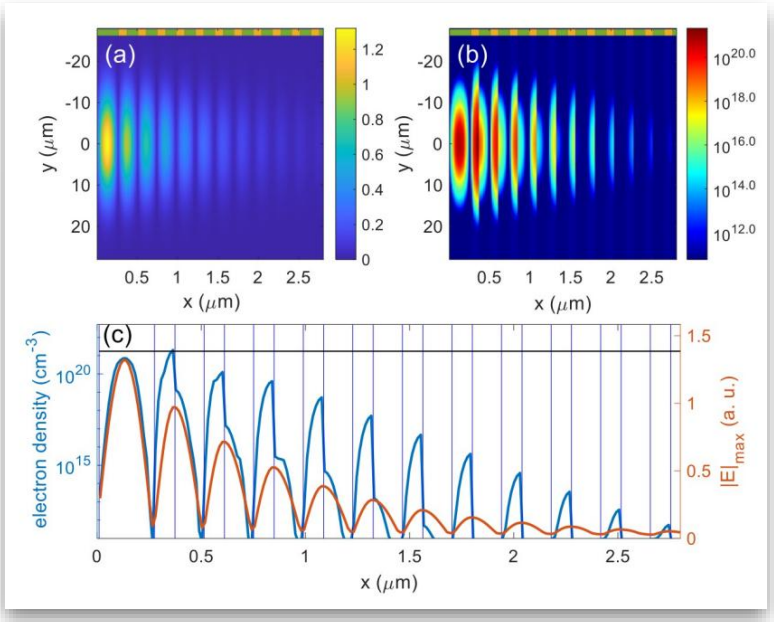
Influence of Defects in Grating [8]

Framework Upgrade ……

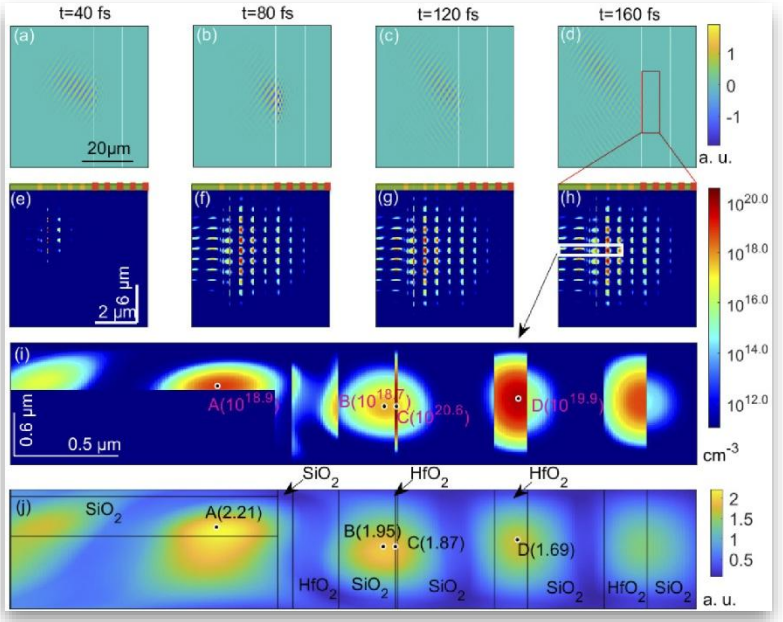
Bulk Fused SiO_2 @ 800nm



MLD Mirror (SiO_2 / HfO_2): 0.7 J/cm², 50 fs @ 800 nm



MLD Grating (SiO_2 / HfO_2 / Ta_2O_5): 0.13 J/cm², 53 fs, *s*-pol @ 1900 nm



[6] S. Zhang, C. Menoni, V. Gruzdev, and E. Chowdhury, "Ultrafast laser material damage simulation—a new look at an old problem," Nanomaterials 12, 1259 (2022)

[7] S. Zhang, et al., "Femtosecond damage experiments and modeling of broadband mid-infrared dielectric diffraction gratings," Opt. Express 29, 39983-39999 (2021)

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2D FDTD Dynamic Ionization Simulation Framework for Ultrashort Laser-Dielectric Interaction [6]

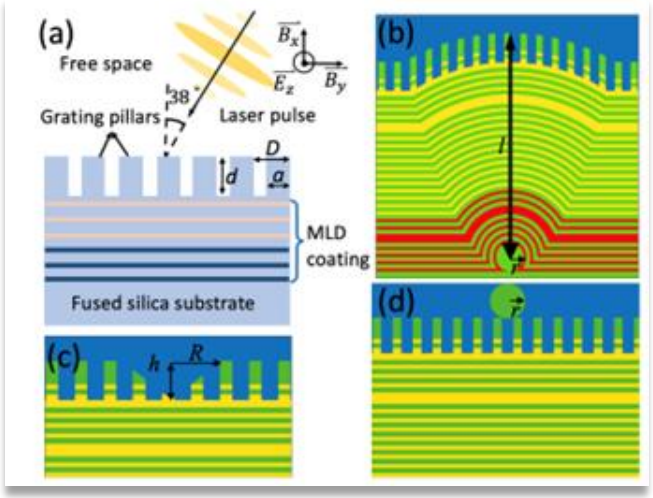
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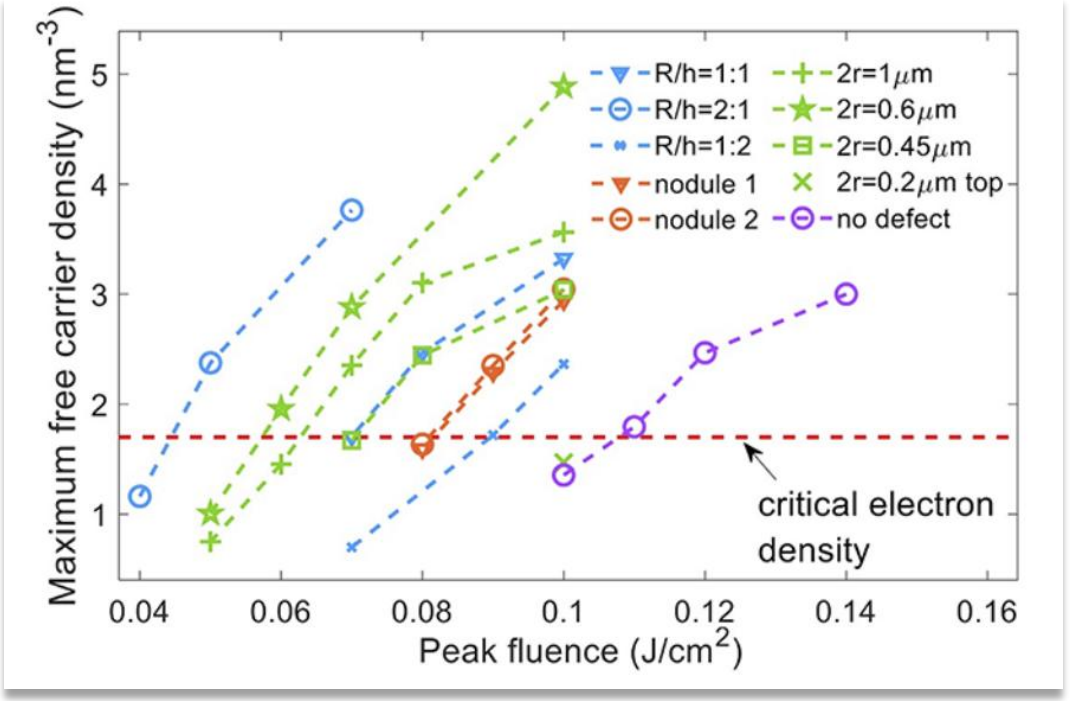
Influence of Defects in Grating [8]

Framework Upgrade ……

(a) No defect
(b) Nodule
(c) Crack
(d) Particle

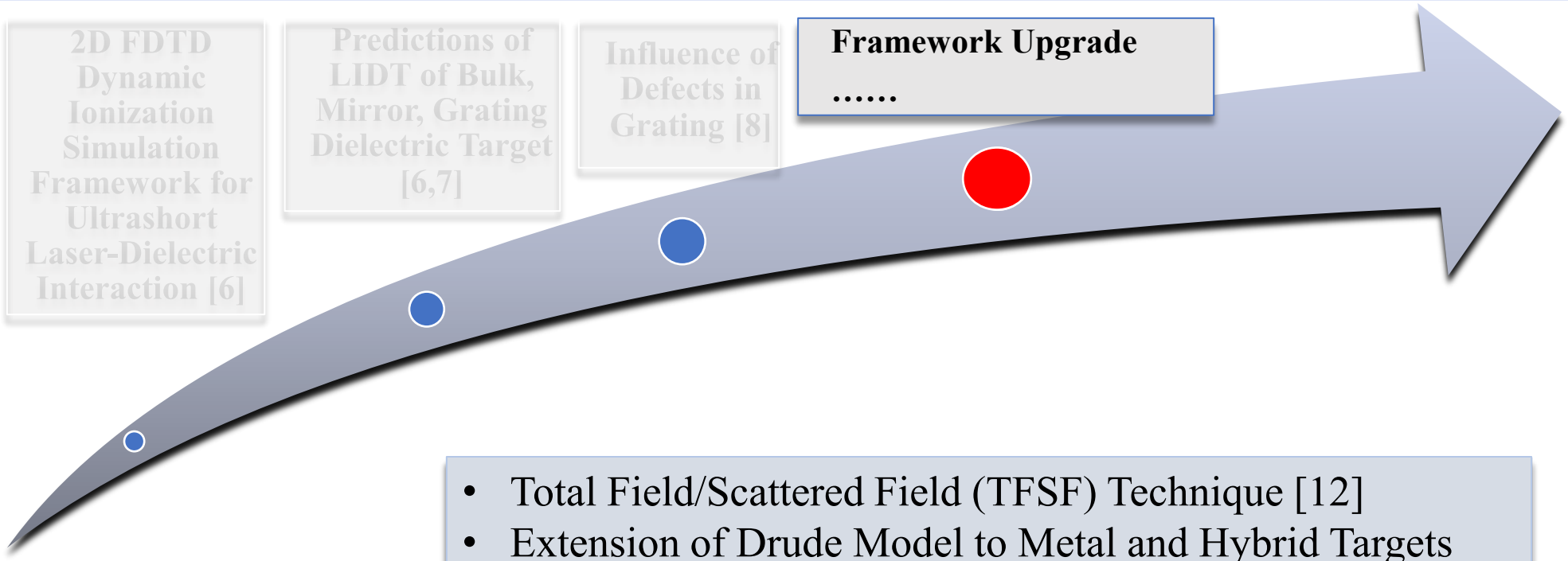
MLD Grating (SiO_2 / HfO_2 / Ta_2O_5):
20 fs, *s*-pol, @ 800 nm





[8] S. Zhang, Z. Su, C. S. Menoni, and E. A. Chowdhury, "Influence of defects on the femtosecond laser damage resistance of multilayer dielectric gratings," Opt. Lett., vol. 48, no. 5, pp. 1212–1215, Mar. 2023. DOI: 10.1364/OL.483581.

Our Efforts



- Total Field/Scattered Field (TFSF) Technique [12]
- Extension of Drude Model to Metal and Hybrid Targets
- Multi Rate Equation (MRE) for Impact Ionization [13]
- ……

[12] Allen Taflove, Susan C. Hagness, "Computational Electrodynamics The Finite-Difference Time-Domain Method" (2000)
[13] B. Rethfeld, "Unified Model for the Free-Electron Avalanche in Laser-Irradiated Dielectrics," Physical Review Letters 92, 187401 (2004). https://doi.org/10.1103/PhysRevLett.92.187401

$$\nu_0(W) = \alpha_0 \left(\frac{W}{W_{ion}} - 1\right)^2, W \geq W_{ion} \quad [14]$$

$$\nu_{col} = \int_0^\infty \nu_0(W) f_e(W)\, dW$$

$$\alpha_0 = \left(\frac{q_e^2}{4\pi\epsilon_0\epsilon_r}\right)^2 \frac{m_e^* A_c^2 A_v^2}{\hbar^3 \left(1 + \frac{2m_e^*}{m_h^*}\right)^{\frac{3}{2}}} \quad [15]$$

W_{ion}: Threshold ionization energy
W: Electron energy
f_e: Electron energy distribution
A_c, A_v: Overlap integrals between the initial and final states of the VB and CB electrons

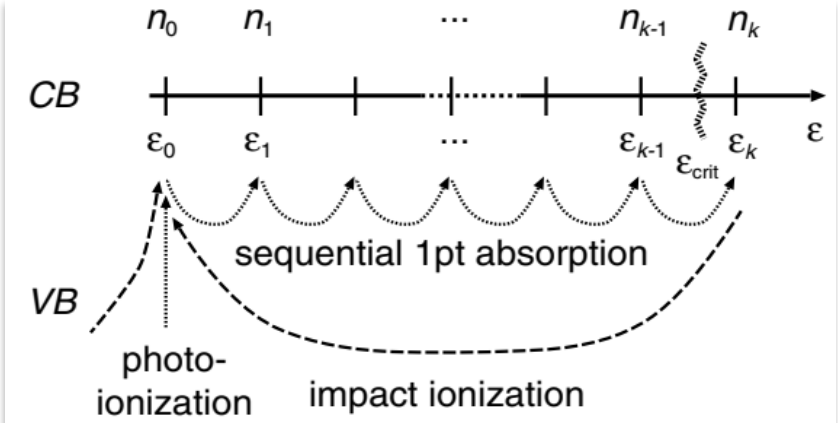


Illustration of the virtual energy ladders of free electrons in the conduction band of a dielectric [13].

$$\omega_{1pt} = \frac{< J_d \cdot E >}{\hbar\omega\, n_e}$$

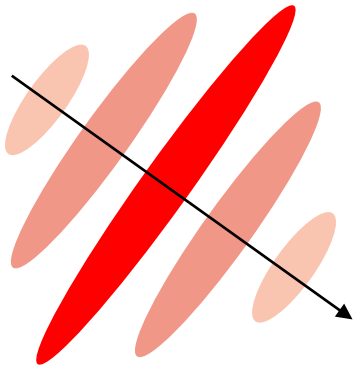
Impact Ionization happens if electrons are in level n_k & $W \geq W_{ion}$

[13] B. Rethfeld, "Unified Model for the Free-Electron Avalanche in Laser-Irradiated Dielectrics," Physical Review Letters 92, 187401 (2004). https://doi.org/10.1103/PhysRevLett.92.187401
[14] L. V. Keldysh, "Kinetic theory of impact ionization in semiconductors," SOVIET PHYSICS JETP, vol. 37, no. 10, 1960.
[15] P.-J. Charpin, et al., Simulation of laser-induced ionization in wide bandgap solid dielectrics with a particle-in-cell code. Optics Express Vol. 32, No. 6, 10175 (2024).

Simulation Set-up

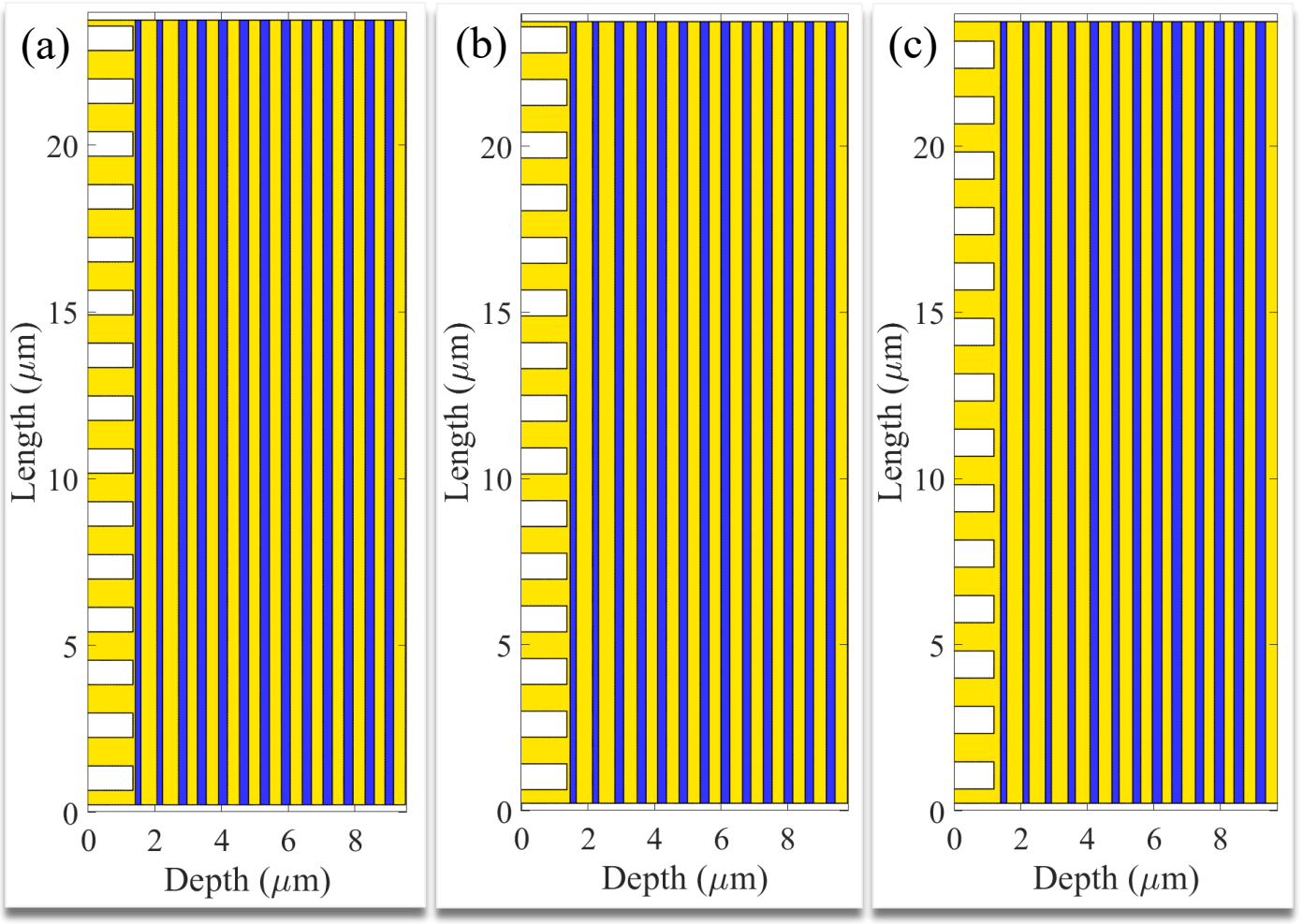
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Source
Wavelength: 1950 nm
FWHM Duration: 70 fs
$1/e^2$ Waist: 4.5 μm
AOI: 36˚~38˚
Polarization: *s*
Single Gaussian Pulse



Grating
Fused SiO_2
Fused HfO_2
27 Alternating Layers
The design in (a) is coated by IBS and IAD for (b,c)

Plymouth Grating LABORATORY



630 l/mm, 54% DC | 630 l/mm, 50% DC | 600 l/mm, 50% DC

Simulation Set-up

Materials Properties [6,7,15]

Material	Bandgap (eV)	m^*_e (a.u.)	m^*_h (a.u.)	n-IAD	n-IBS	Mass Density (g/cc)	Ac, Av	τ_{re} (fs)	Z
Fused SiO_2	9	0.6	8	1.4500	1.4792	2.20	0.104	220	4
Fused HfO_2	5.7	1.09	1.12	1.9220	1.9954	9.68	0.104	1050	4

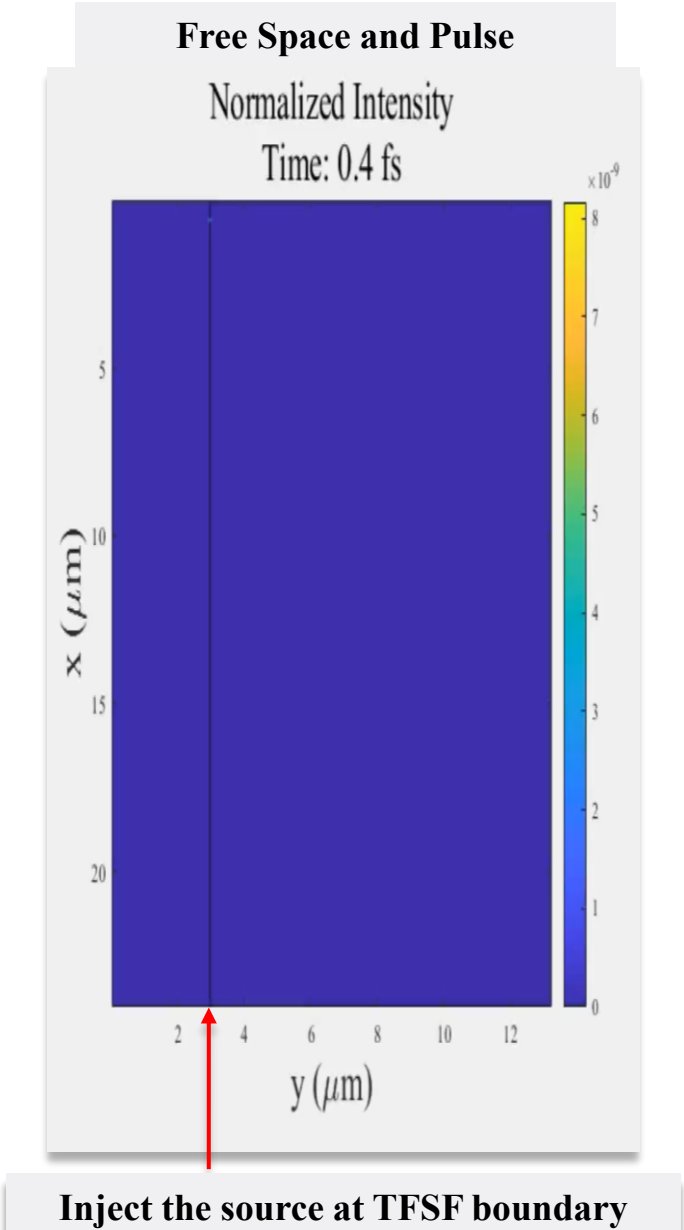
Provided by manufacturer at 1.94 µm

Some Assumptions

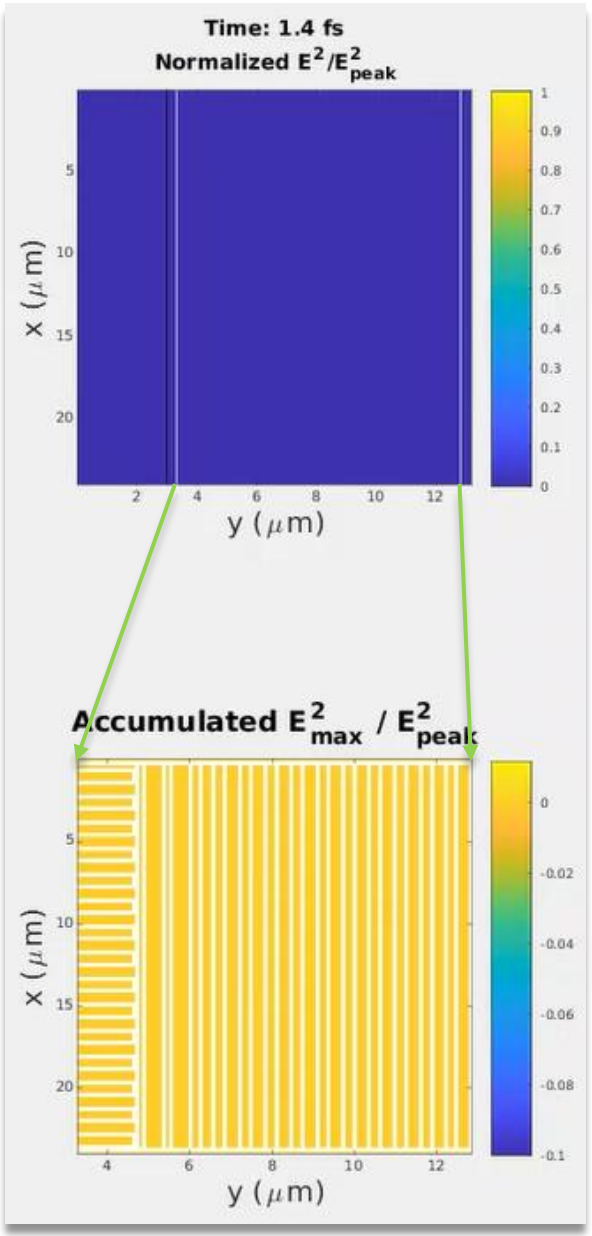
- Density may differ from bulk; effective mass hard to measure and vendor data absent.
- Overlap integrals are assumed to be the same with SiO_2 for HfO_2 (no data).
- There are four covalent bonds per molecule, and each molecule can only contribute one electron.

[6] S. Zhang, C. Menoni, V. Gruzdev, and E. Chowdhury, "Ultrafast laser material damage simulation—a new look at an old problem," Nanomaterials 12, 1259 (2022)
[7] S. Zhang, et al., "Femtosecond damage experiments and modeling of broadband mid-infrared dielectric diffraction gratings," Opt. Express 29, 39983-39999 (2021)
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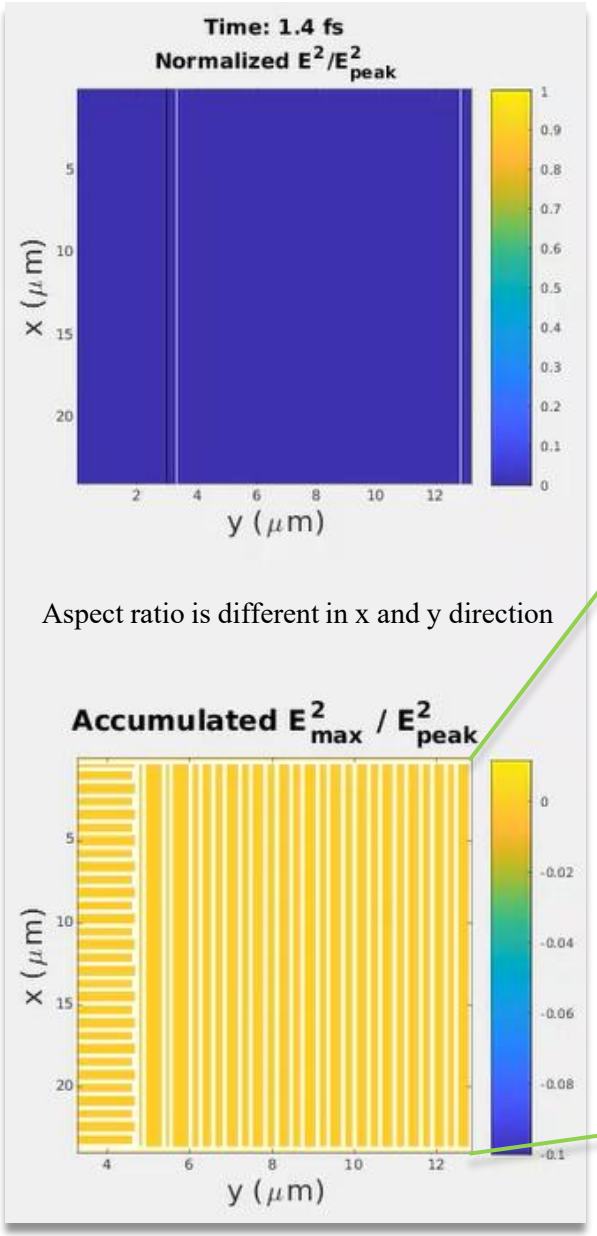
Results (a) 630 l/mm, 54% DC (IBS), 0.45 J/cm²

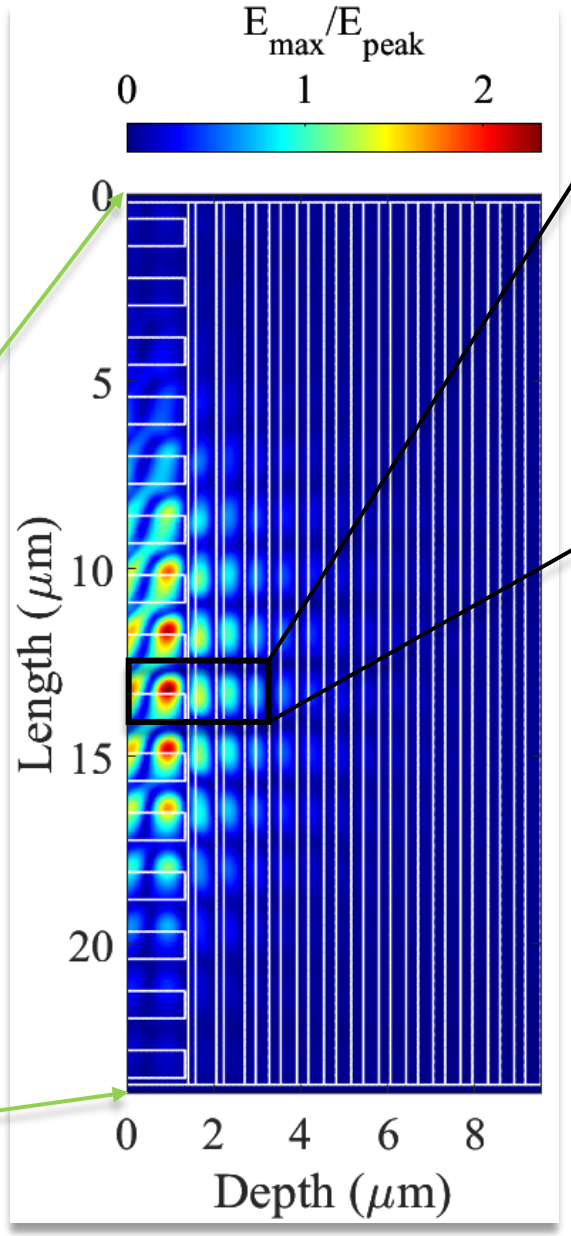


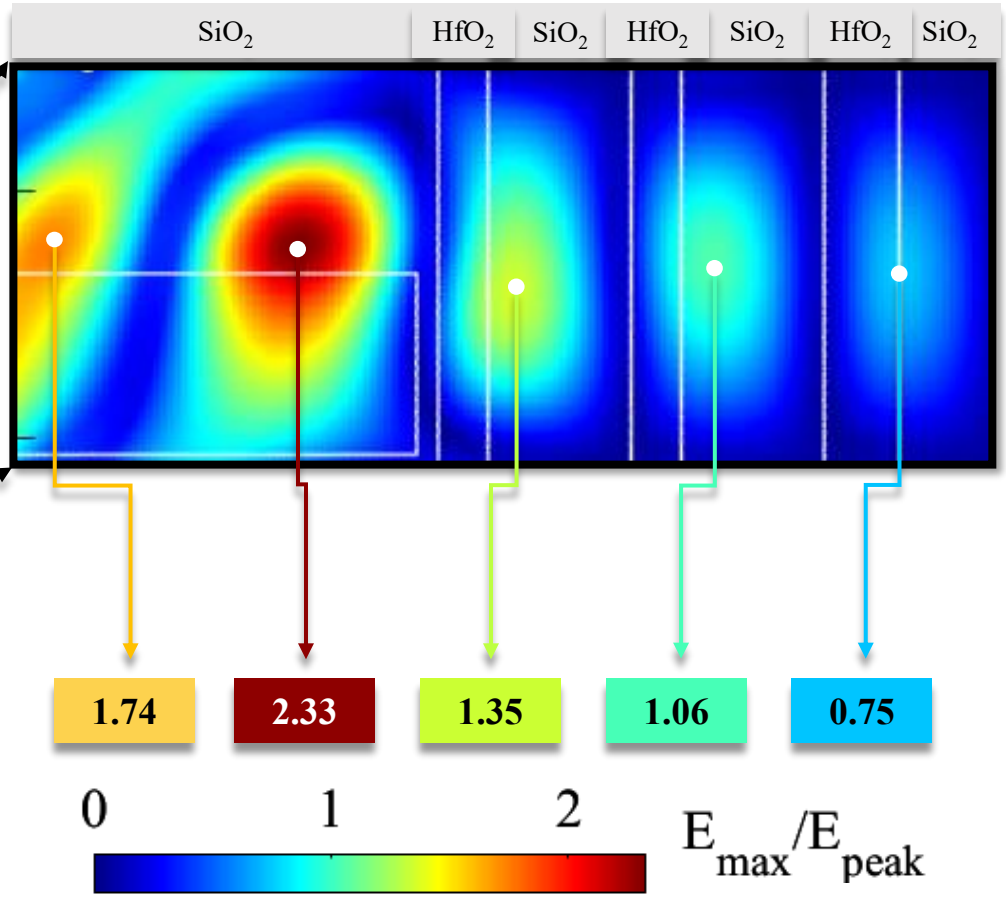
Add Target



Results (a) 630 l/mm, 54% DC (IBS), 0.45 J/cm²

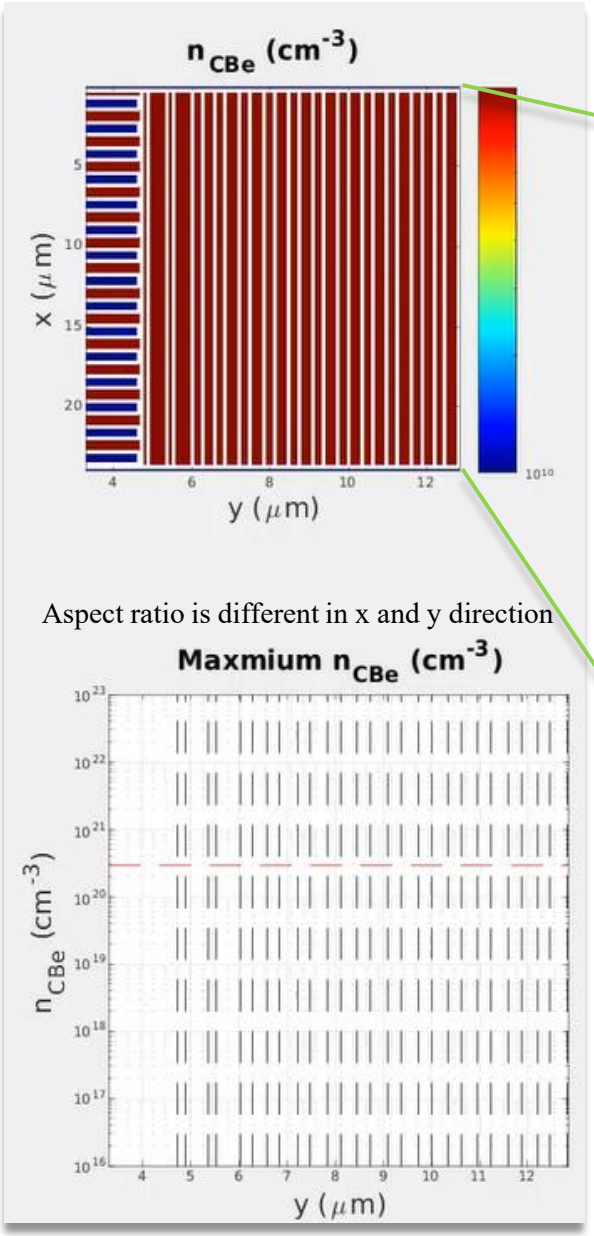


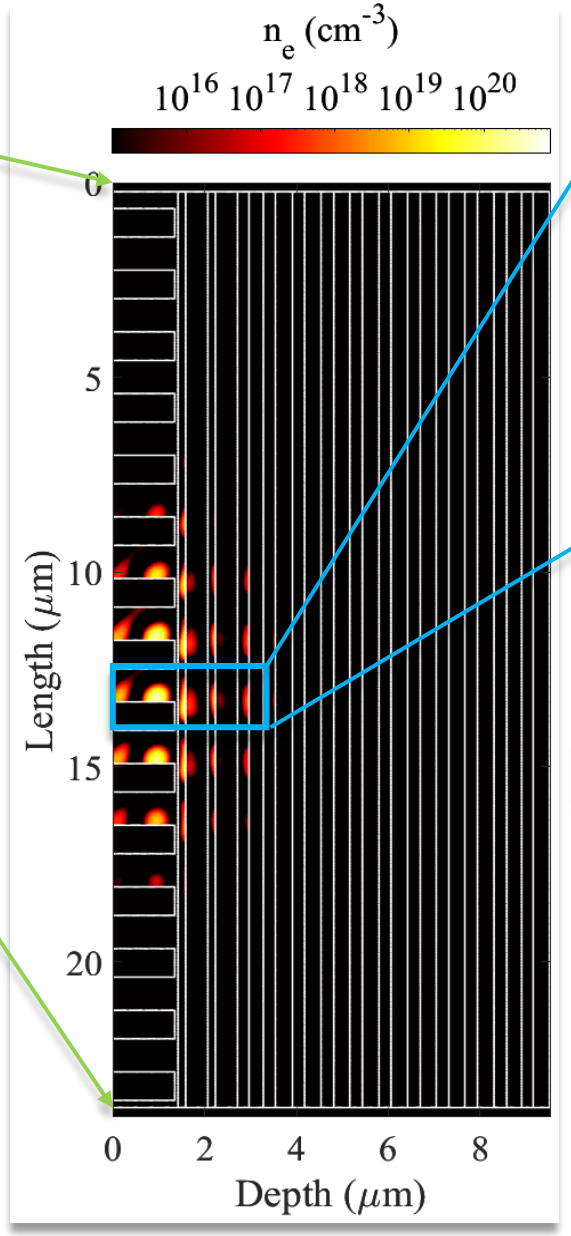


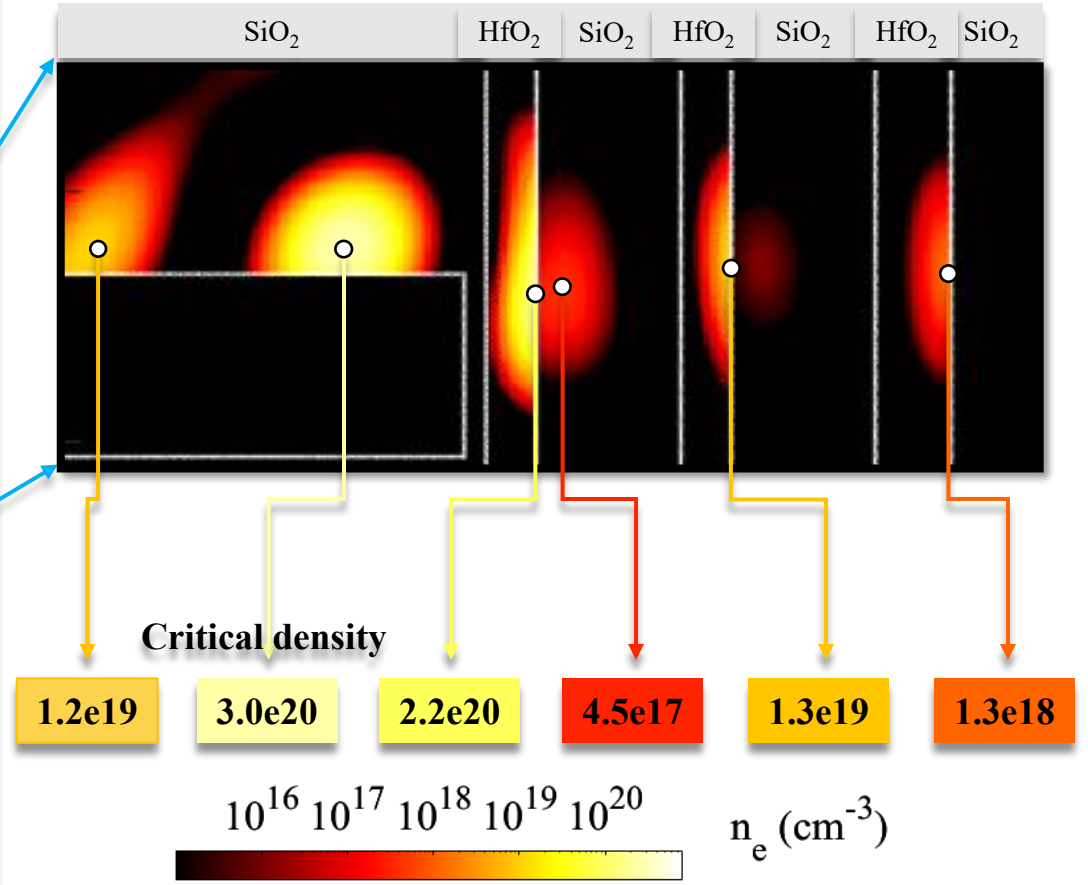


E_{max}: Maximum accumulated electric field
E_{peak}: Peak electric field in free space

Results (a) 630 l/mm, 54% DC (IBS), 0.45 J/cm²

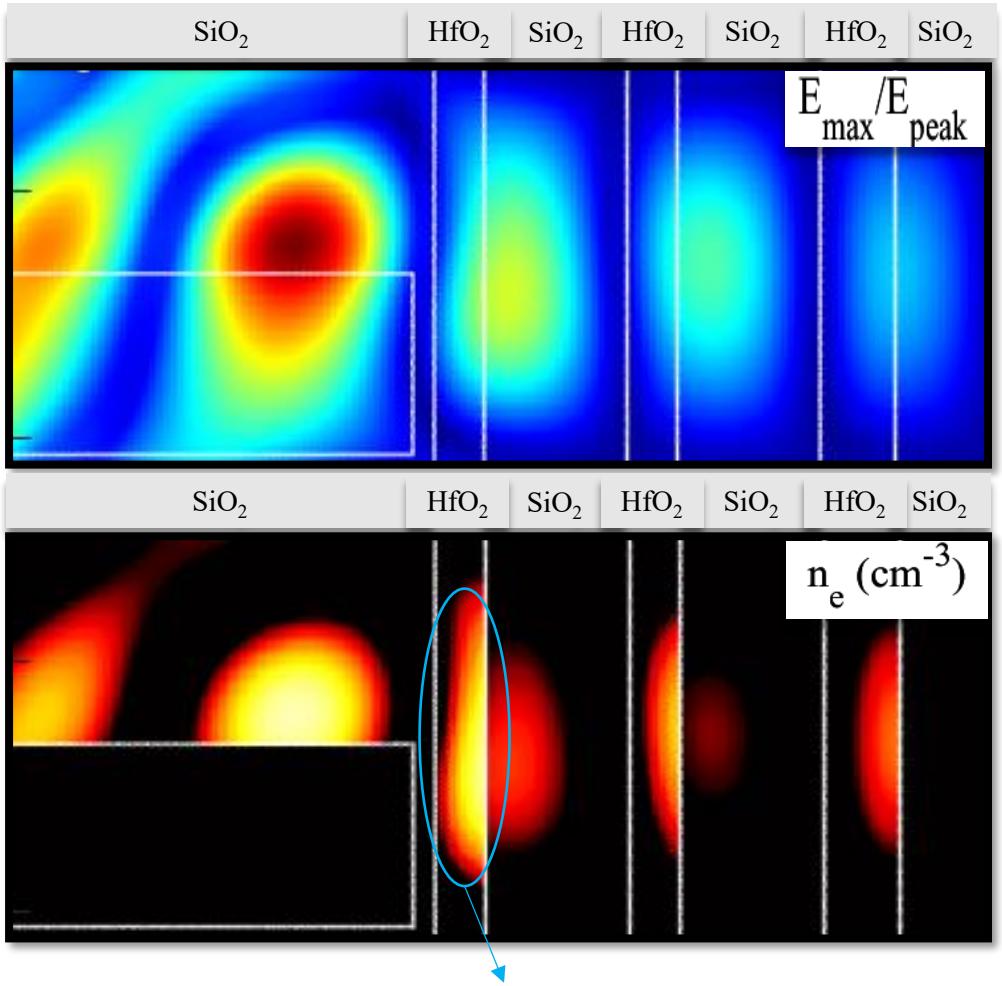






Damage may initiate from the bottom of the pillar in SiO_2 layer !

Results (a) 630 l/mm, 54% DC (IBS), 0.45 J/cm²



May damage too, but cannot be predicted from the field analysis alone

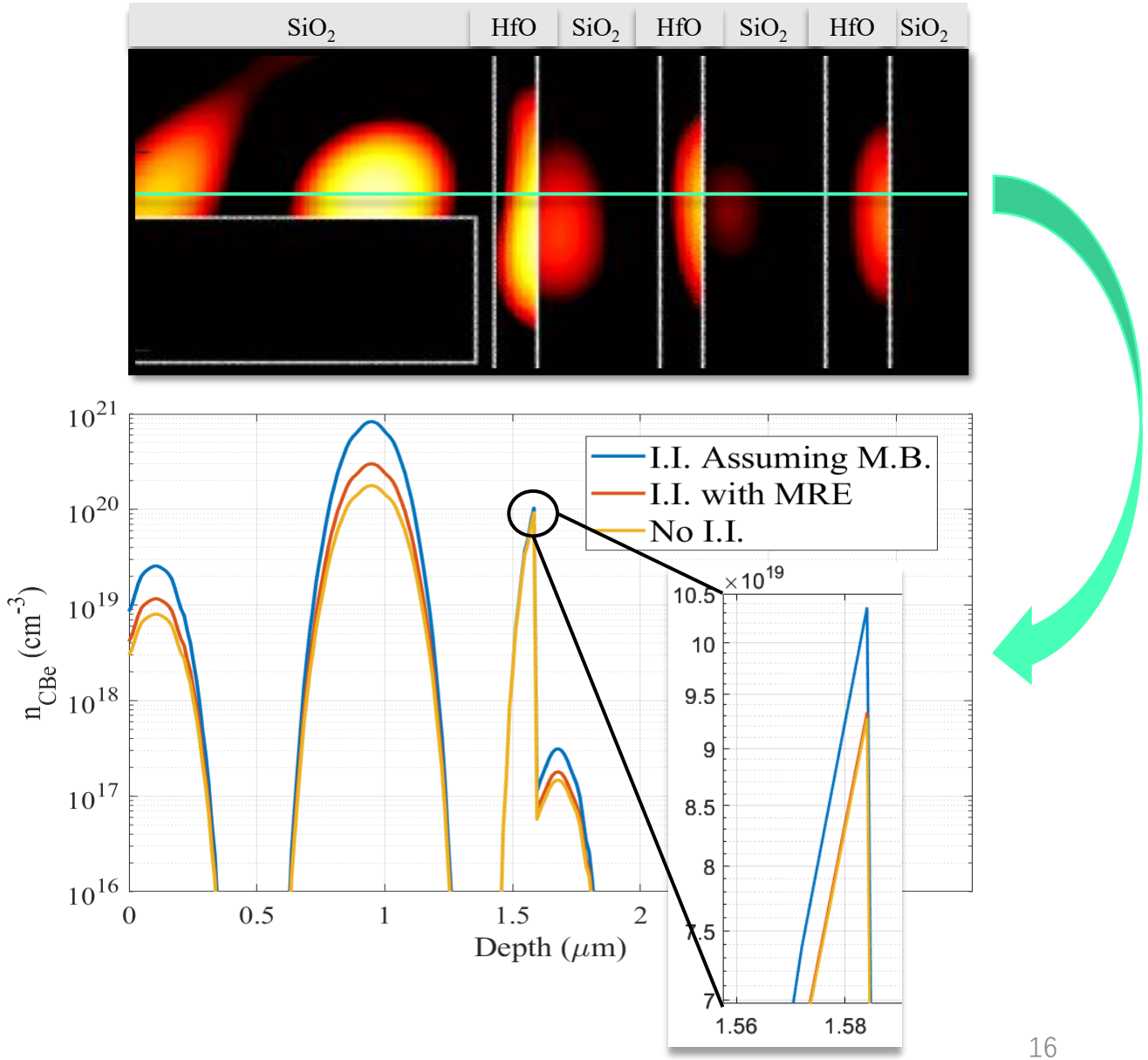
- The maximum E-field enhancement coincides with the global peak electron density in this case, but the first HfO_2 layer may still damage as its density is similar to neighboring layers.
- Maximum E-field ≠ peak density; analyzing field enhancement alone is not sufficient to comprehensively evaluate LIDT.

Results (a) 630 l/mm, 54% DC (IBS), 0.45 J/cm²

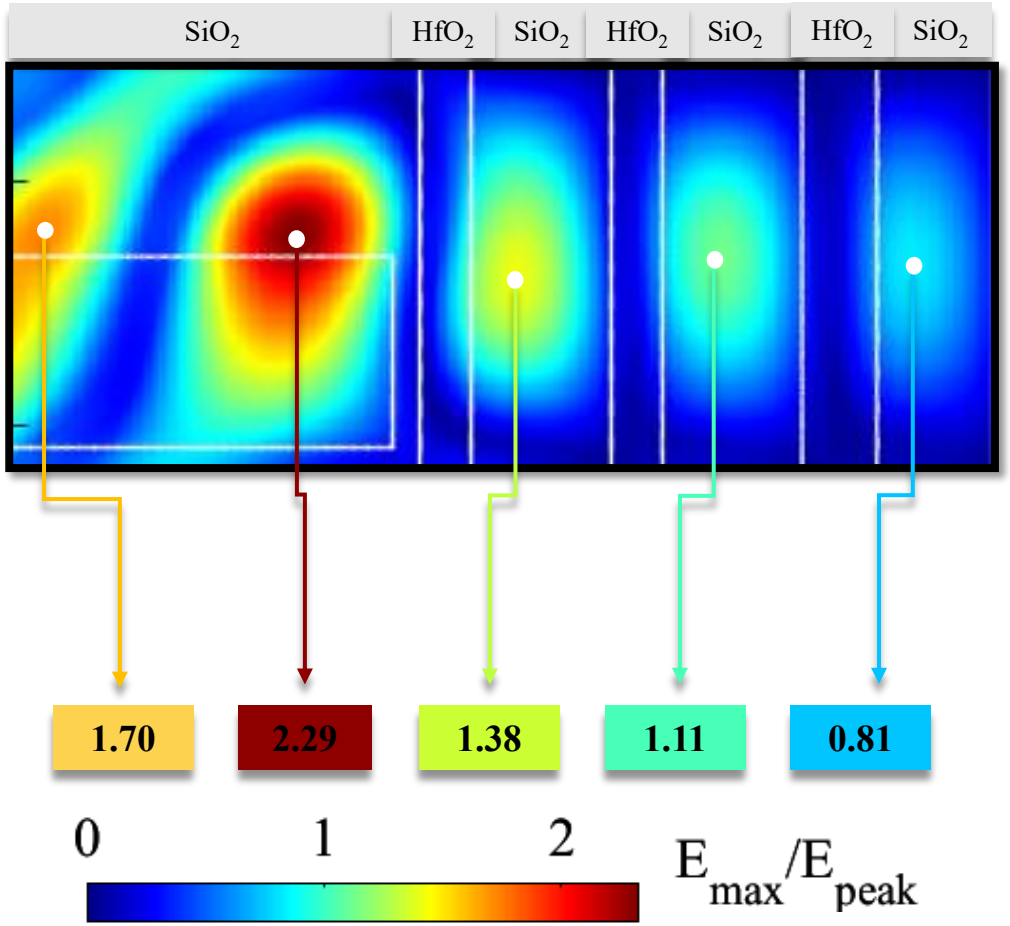
- About an order-of-magnitude higher electron density is observed at the SiO_2 pillar base between No I.I. and I.I. assuming M.B cases.

Why does HfO_2 not change too much?

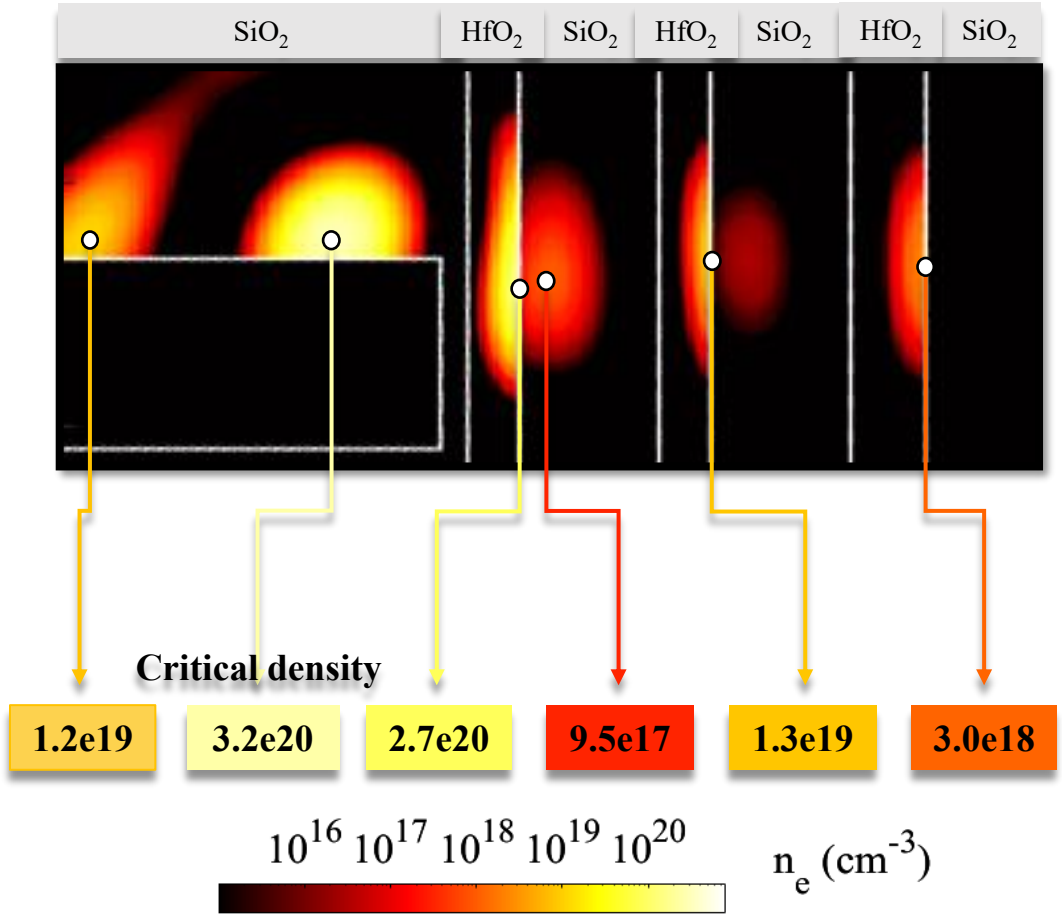
- Higher n refractive inde
  → weaker internal field
- Larger m_e^*
  → lower energy absorption per electron
  → smaller α_0.



Results (b) 630 l/mm, 50% DC, 0.47 J/cm²

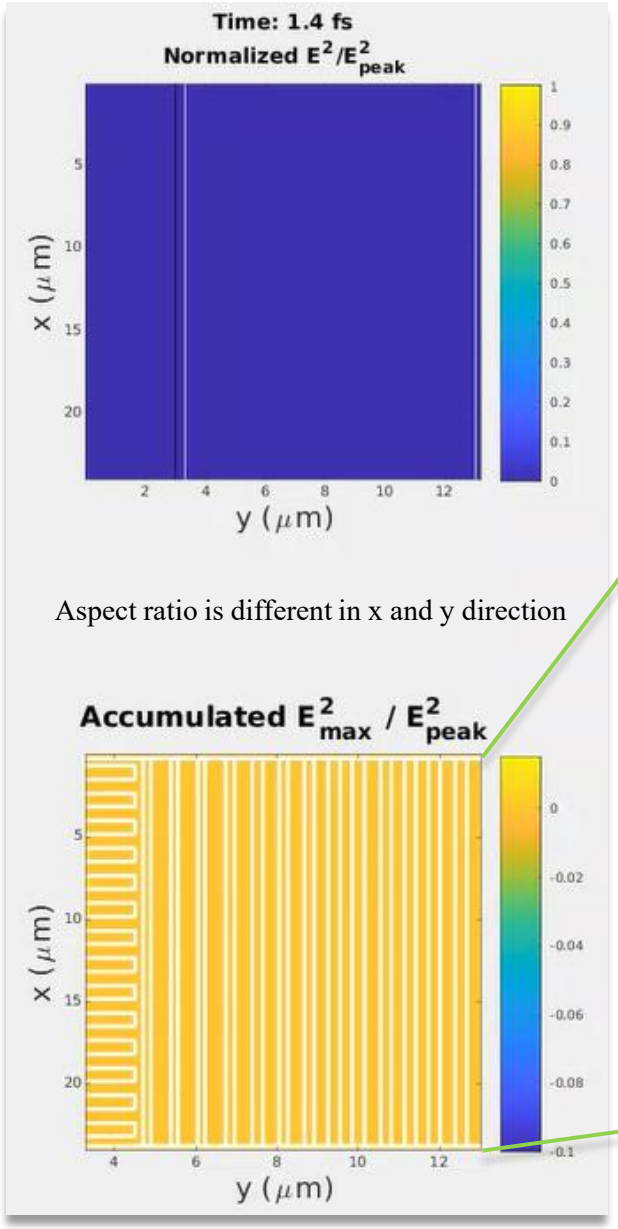


E_{max}: Maximum accumulated electric field
E_{peak}: Peak electric field in free space

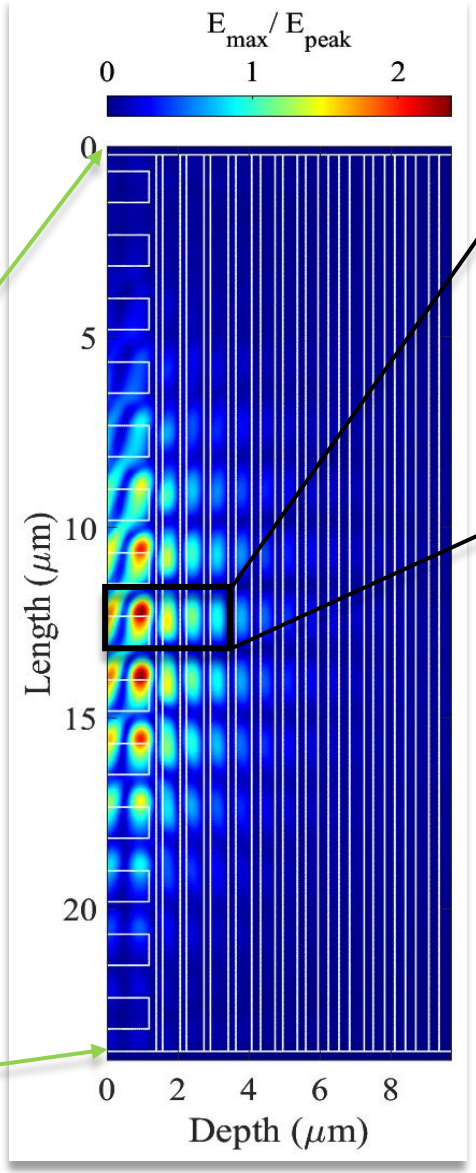


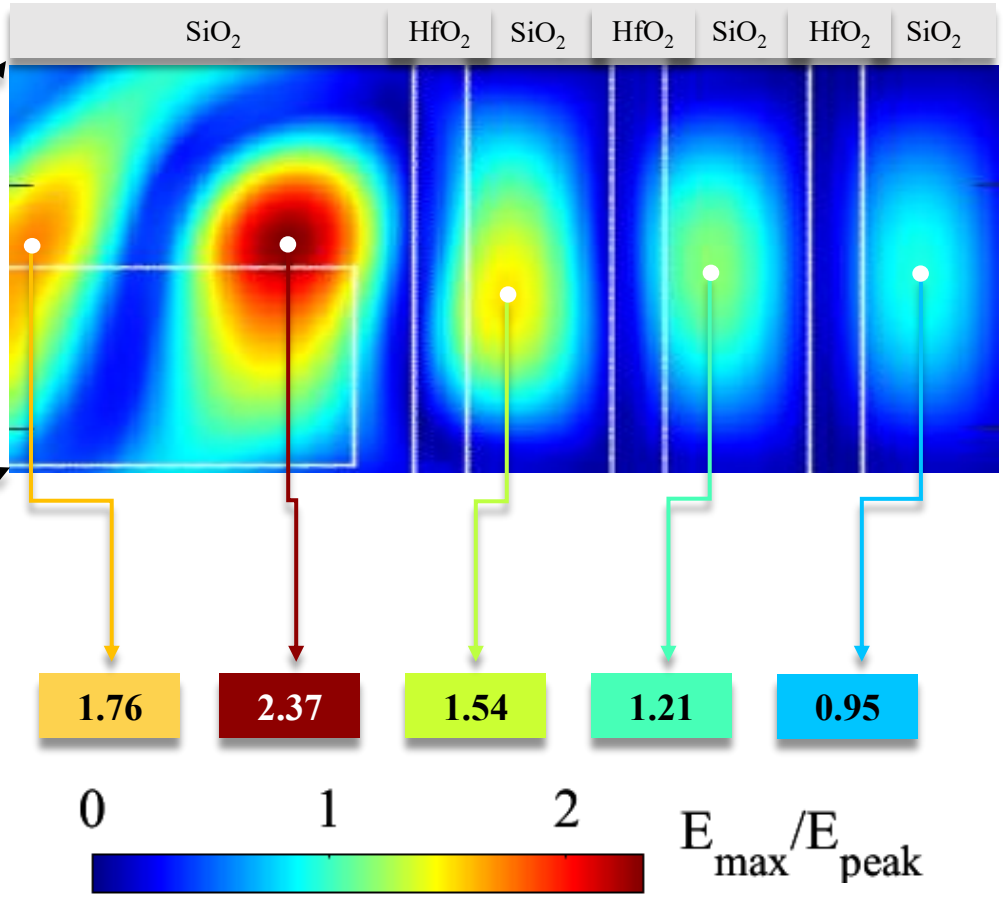
Damage may initiate from the bottom of the pillar in SiO_2 layer !

Results (c) 600 l/mm, 50% DC, 0.40 J/cm²



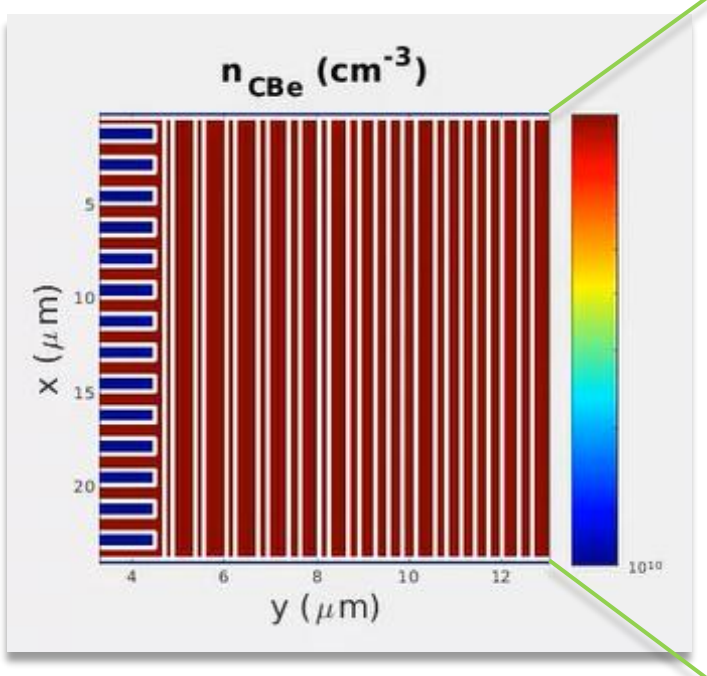
Aspect ratio is different in x and y direction



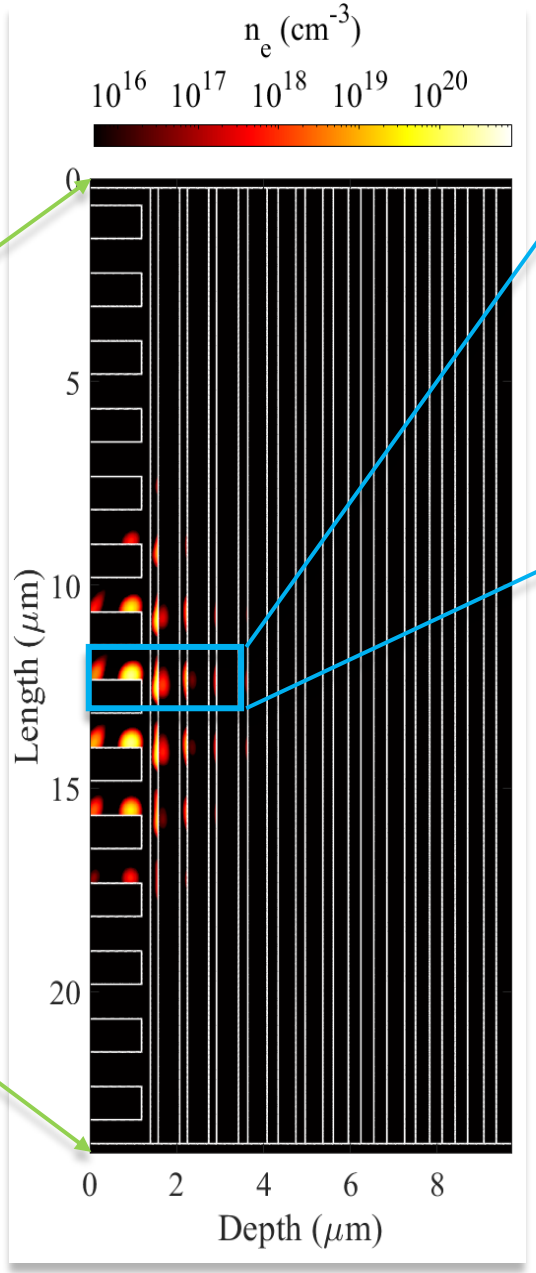


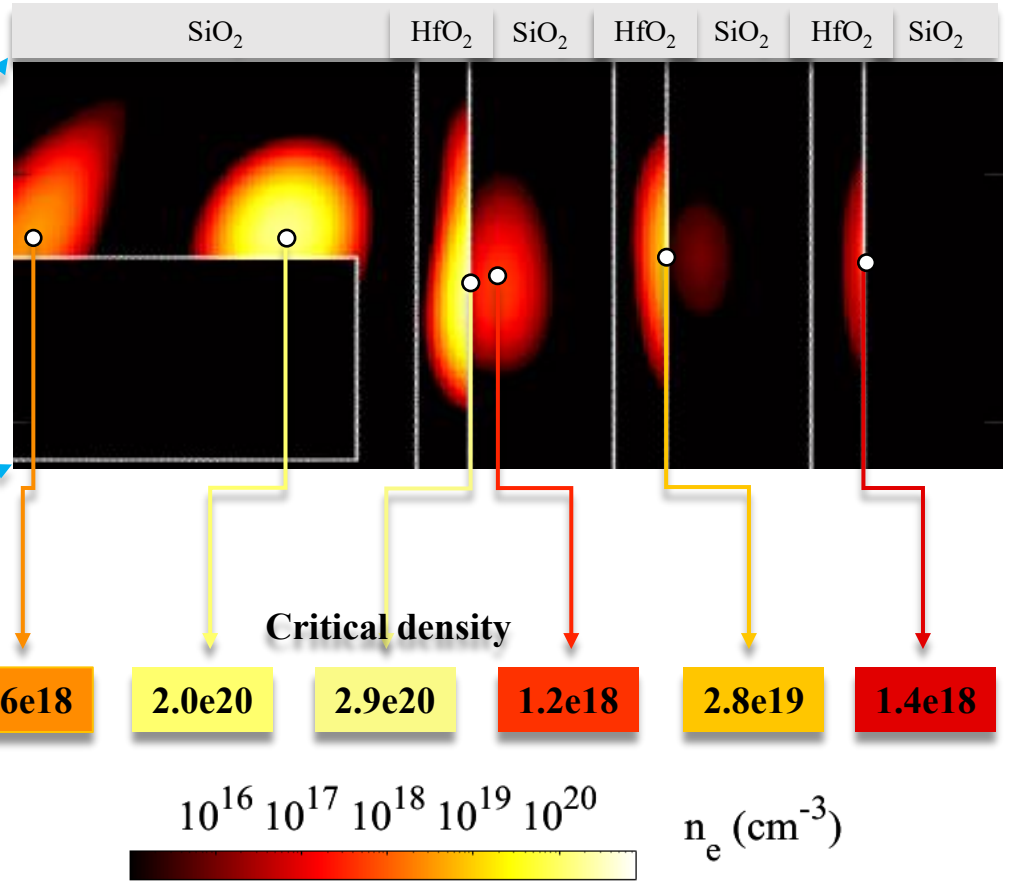
E_{max}: Maximum accumulated electric field
E_{peak}: Peak electric field in free space

Results (c) 600 l/mm, 50% DC, 0.40 J/cm²



Aspect ratio is different in x and y direction



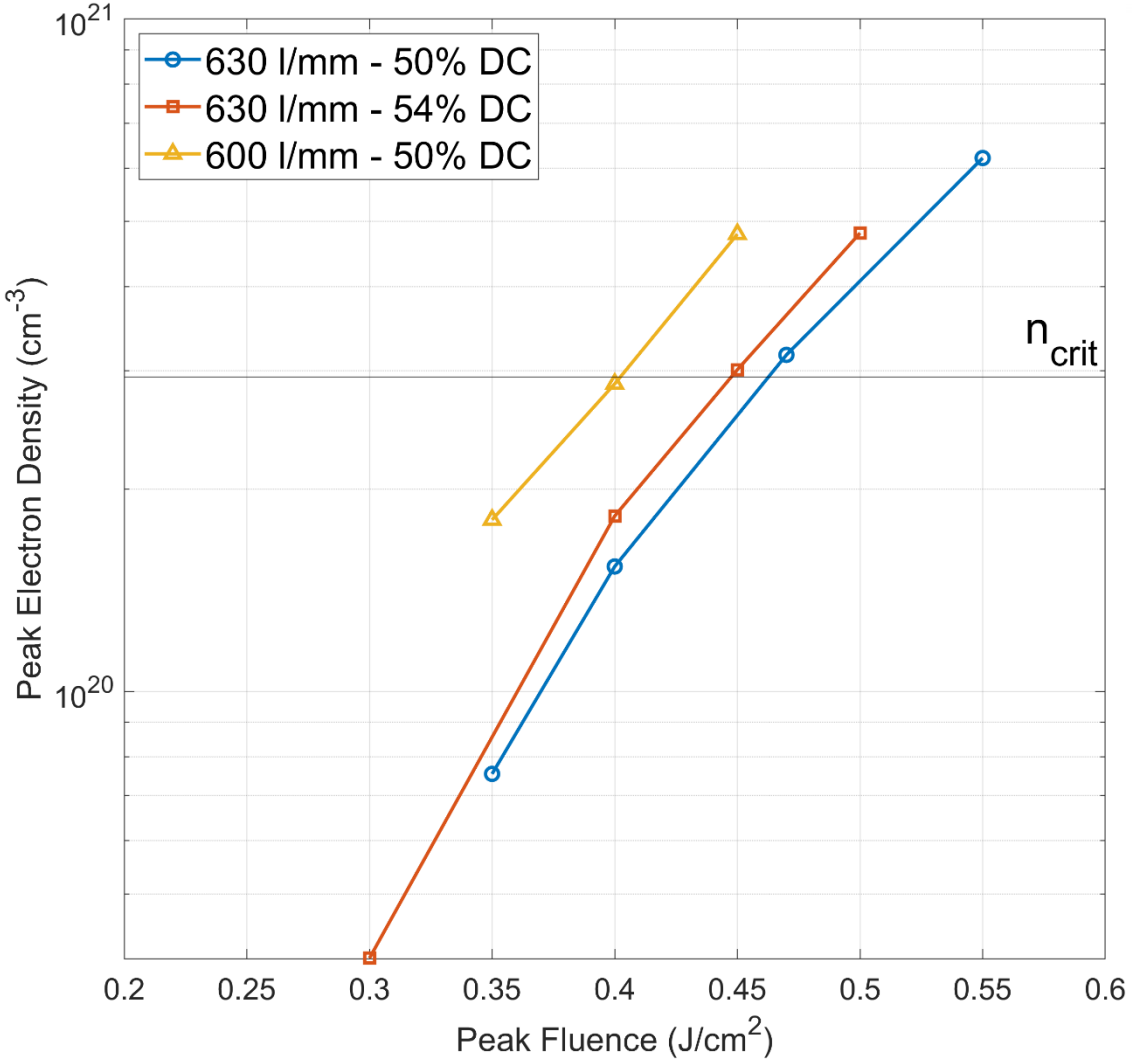


Damage may initiate from the first HfO_2 layer !

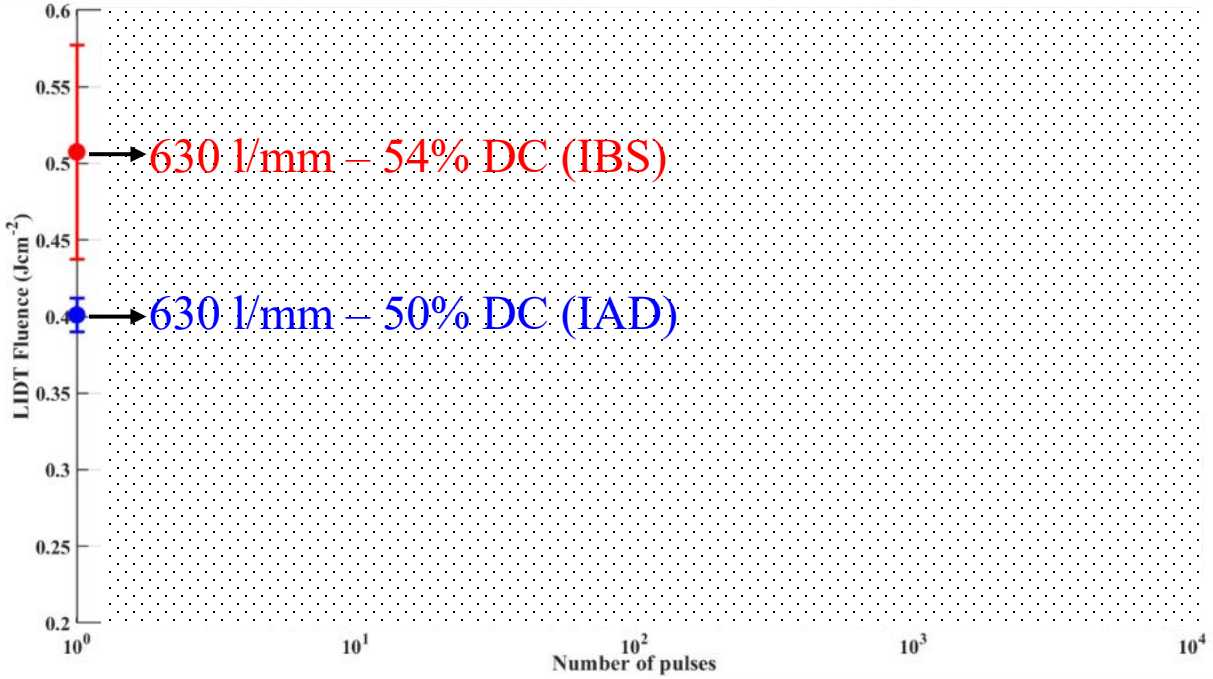
Results

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Simulation Predicted LIDT for 3 Grating Designs



Experiment of S on 1 Damage Test



To unlock the rest of the data, please check Yaseen's poster #13729-72 tomorrow

Summary

- Implemented Keldysh photoionization model with cycle-averaged approach and MRE for the impact ionization.
- Peak field strength may not correspond to the global peak electron density in multilayer targets as the electronic properties are different.
- Predicted well matched LIDT for the gratings which has 630 l/mm at 1950nm with the experiments.

Design	Deposition	Predicted LIDT	Tested LIDT	Error	Initiate Damage Site
(a) 630 l/mm, 54% DC	IBS	0.45 J cm^{-2}	0.51±0.06 J cm^{-2}	11.8%	First SiO_2 layer
(b) 630 l/mm, 50% DC	IAD	0.46 J cm^{-2}	0.40±0.01 J cm^{-2}	15.0%	First SiO_2 layer
(c) 600 l/mm, 50% DC	IAD	0.40 J cm^{-2}	/	/	First HfO_2 layer

Thank you

Bibliography

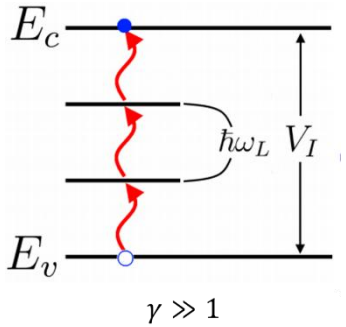
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[15] P.-J. Charpin, et al., Simulation of laser-induced ionization in wide bandgap solid dielectrics with a particle-in-cell code. Optics Express Vol. 32, No. 6, 10175 (2024).

Back-up

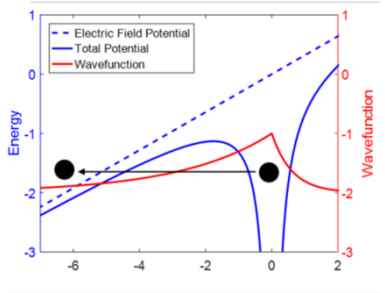
Photoionization[1]

$$\gamma = \frac{\omega\sqrt{m_e W_0}}{eE}$$

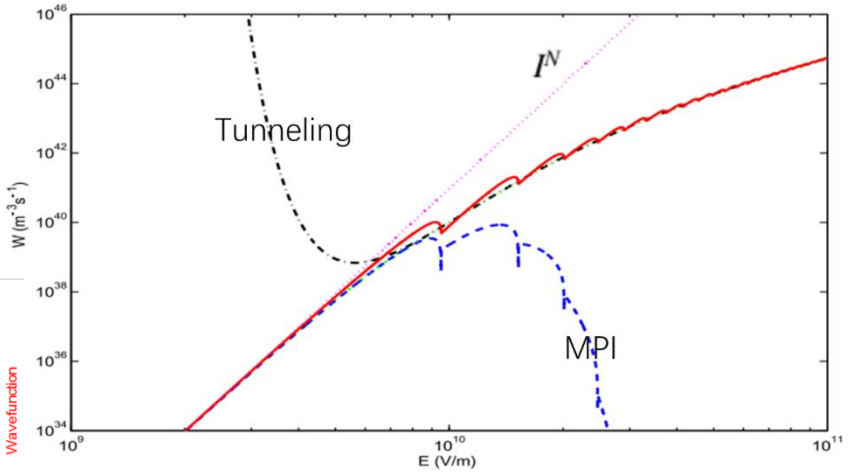
W_0: Ionization energy
E: External electric field
ω : Laser frequency



$\gamma \gg 1$



$\gamma \ll 1$



Left: Multi-photon/Tunneling Ionization Cartoon [2]. **Right:** Strong Field Ionization Rates Based on Different Models (wavelength 800 nm, bandgap 6.5 eV) [3]

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Defining the density n_j as the density of electrons at energy ε_j, one can describe this process with the following system of rate equations, representing the multiple rate equation (MRE):

$$\begin{aligned}
\dot{n}_0 &= \dot{n}_{\mathrm{pi}} + 2\,\tilde{\alpha}\,n_k - W_{1pt}(\varepsilon_0)\,n_0,\\
\dot{n}_1 &= W_{1\mathrm{pt}}(\varepsilon_0)\,n_0 - W_{1\mathrm{pt}}(\varepsilon_1)\,n_1,\\
&\vdots\\
\dot{n}_{k-1} &= W_{1\mathrm{pt}}(\varepsilon_{k-2})\,n_{k-2} - W_{1\mathrm{pt}}(\varepsilon_{k-1})\,n_{k-1},\\
\dot{n}_k &= W_{1\mathrm{pt}}(\varepsilon_{k-1})\,n_{k-1} - \tilde{\alpha}\,n_k, \qquad \text{with } k = \left\lceil \frac{\varepsilon_{\mathrm{crit}}}{\hbar\omega_L} \right\rceil.
\end{aligned} \tag{2}$$

Adding up these equations yields

$$\frac{dn_{\mathrm{total}}}{dt} = \dot{n}_{\mathrm{pi}}(E_L) + \tilde{\alpha}\,n_k, \tag{3}$$

with $n_{\mathrm{total}} = \sum_{j=1}^{k} n_j$. The difference in the last term of Eq. (3) compared to Eq. (1) is substantial: while in Eq. (1)

For 1950nm, there are about 14 virtual levels in code

```
rise = is_dfilm; %& wcolfilm>0;
nCBefilm0(is_dfilm) = nCBefilm0(is_dfilm) + nCBefilm(is_dfilm)-nCBefilm_r(is_dfilm) +...
    dt*wcolfilm(rise).*nCBefilm14(rise) - dt*w1pt(rise).* nCBefilm0(rise);

nCBefilm1(rise) = nCBefilm1(rise) + dt*w1pt(rise).* (nCBefilm0(rise) - nCBefilm1(rise));
nCBefilm2(rise) = nCBefilm2(rise) + dt*w1pt(rise).* (nCBefilm1(rise) - nCBefilm2(rise));
nCBefilm3(rise) = nCBefilm3(rise) + dt*w1pt(rise).* (nCBefilm2(rise) - nCBefilm3(rise));
nCBefilm4(rise) = nCBefilm4(rise) + dt*w1pt(rise).* (nCBefilm3(rise) - nCBefilm4(rise));
nCBefilm5(rise) = nCBefilm5(rise) + dt*w1pt(rise).* (nCBefilm4(rise) - nCBefilm5(rise));
nCBefilm6(rise) = nCBefilm6(rise) + dt*w1pt(rise).* (nCBefilm5(rise) - nCBefilm6(rise));
nCBefilm7(rise) = nCBefilm7(rise) + dt*w1pt(rise).* (nCBefilm6(rise) - nCBefilm7(rise));
nCBefilm8(rise) = nCBefilm8(rise) + dt*w1pt(rise).* (nCBefilm7(rise) - nCBefilm8(rise));
nCBefilm9(rise) = nCBefilm9(rise) + dt*w1pt(rise).* (nCBefilm8(rise) - nCBefilm9(rise));
nCBefilm10(rise) = nCBefilm10(rise) + dt*w1pt(rise).* (nCBefilm9(rise) - nCBefilm10(rise));
nCBefilm11(rise) = nCBefilm11(rise) + dt*w1pt(rise).* (nCBefilm10(rise) - nCBefilm11(rise));
nCBefilm12(rise) = nCBefilm12(rise) + dt*w1pt(rise).* (nCBefilm11(rise) - nCBefilm12(rise));
nCBefilm13(rise) = nCBefilm13(rise) + dt*w1pt(rise).* (nCBefilm12(rise) - nCBefilm13(rise));
nCBefilm14(rise) = nCBefilm14(rise) + dt*w1pt(rise).*nCBefilm13(rise) - dt*wcolfilm(rise).*nCBefilm14(rise);
```

```
JExfilm = Jdxfilm.*Exfilm;
JEyfilm = Jdyfilm.*Eyfilm;
JEzfilm = Jdzfilm.*Ezfilm;
Q = dt*abs(JExfilm+JEyfilm+JEzfilm);
Qdot = Q/(dt*10^-15)/1e12;
w1pt(is_dfilm) = Qdot(is_dfilm)./nCBefilm(is_dfilm)/ (1e15*1.054e-34*omega)*10^-15;  % [1/fs]
```

$$W_{1\gamma} = \frac{\langle \mathbf{J}_d \cdot \mathbf{E} \rangle}{n_e\,\hbar\omega}$$

Results (a) 630 l/mm, 54% DC (IBS), 0.45 J/cm²

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Electron Dynamics

ω_{pho}

ω_{col}

